

11.1 Classwork: Quadratics graphing

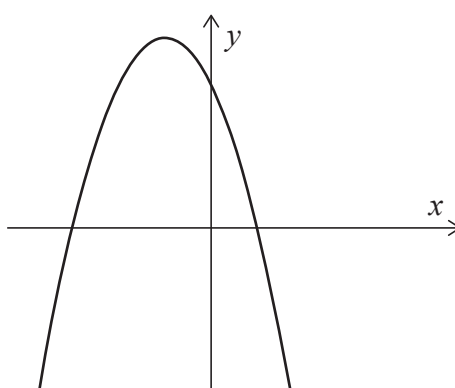
Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 7]

Consider the function $f(x) = -2(x - 1)(x + 3)$, for $x \in \mathbb{R}$. The following diagram shows part of the graph of f .



(a) For the graph of f

(i) find the x -coordinates of the x -intercepts;

(ii) find the coordinates of the vertex.

[5]

The function f can be written in the form $f(x) = -2(x - h)^2 + k$.

(b) Write down the value of h and the value of k .

[2]

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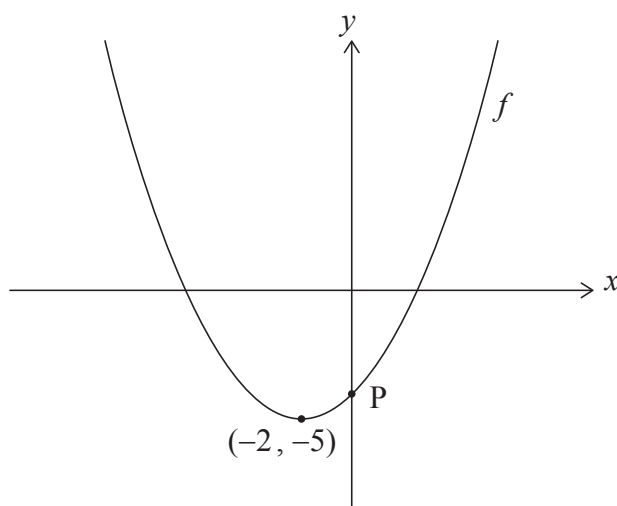
Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

7. [Maximum mark: 17]

The following diagram shows part of the graph of a quadratic function f .

The vertex of the parabola is $(-2, -5)$ and the y -intercept is at point P.



(a) Write down the equation of the axis of symmetry.

[1]

The function can be written in the form $f(x) = \frac{1}{4}(x-h)^2 + k$, where $h, k \in \mathbb{Z}$.

(b) Write down the values of h and k .

[2]

(c) Find the y -coordinate of P.

[2]

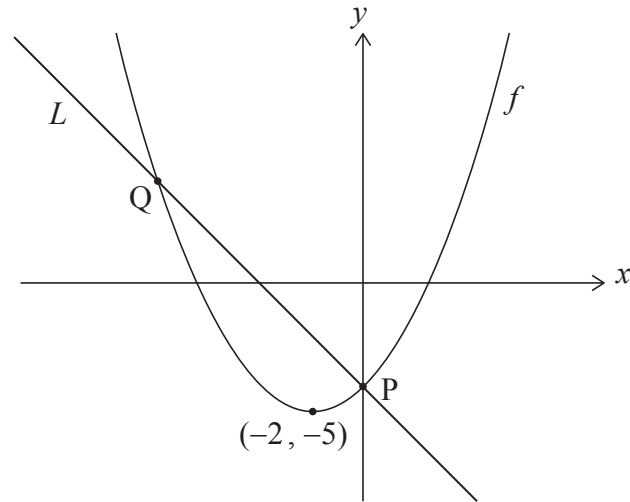
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(Question 7 continued)

In the following diagram, the line L is normal to the graph of f at point P .



- (d) Find the equation of the line L , in the form $y = ax + b$. [4]

The line L intersects the graph of f at a second point, Q , as shown above.

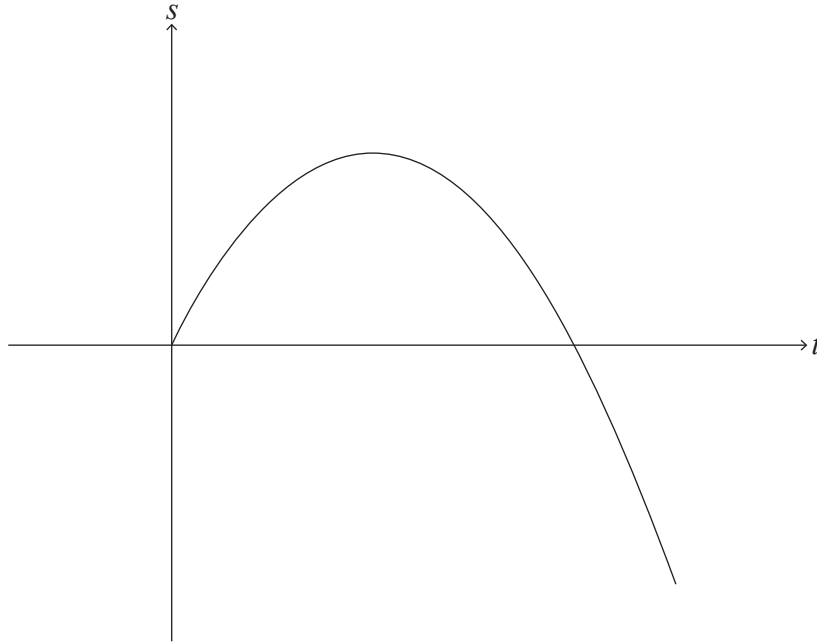
- (e) Calculate the distance between P and Q . [8]



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9. [Maximum mark: 14]

Particle A travels in a straight line such that its displacement, s metres, from a fixed origin after t seconds is given by $s(t) = 8t - t^2$, for $0 \leq t \leq 10$, as shown in the following diagram.



Particle A starts at the origin and passes through the origin again when $t = p$.

(a) Find the value of p . [2]

Particle A changes direction when $t = q$.

(b) (i) Find the value of q .
(ii) Find the displacement of particle A from the origin when $t = q$. [4]

(c) Find the distance of particle A from the origin when $t = 10$. [2]

The total distance travelled by particle A is given by d .

(d) Find the value of d . [2]

A second particle, particle B, travels along the same straight line such that its velocity is given by $v(t) = 14 - 2t$, for $t \geq 0$.

When $t = k$, the distance travelled by particle B is equal to d .

(e) Find the value of k . [4]

References:

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